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Cover Letter: Helix AI Engineer, Robot Learning

Figure San Jose 2026-07-08

I do VLA manipulation research nights and weekends with Merlyn Labs, the 3-person collective I co-founded, and Helix is exactly the type of thing that work has been heading towards. And now that Figure 03s are running in real workstations, the fleet's deployment data holds contextual learning no benchmark can give you, and mining it is the problem I'd most like to work on.

By day I'm an AI Research Engineer at BMO's AI Centre of Excellence, where I built a deterministic eval that caught a GenAI tool systematically downplaying investment risk on $200B+ AUM. I'm also partway through an M.Eng in AI & Robotics at the University of Toronto. The through-line in my work is getting in the mud and finding where learned systems break, then fixing them.

- **Identified proprioceptive collapse as a critical VLA failure mode:,** masking 60% of proprioception improved task success by up to 48% in Stanford's BEHAVIOR-1K Challenge (8th place, standard track, trained on 10,000+ demonstrations across 22 tasks).
- **Doubled π0.5's position-swap success on LIBERO-PRO (21% to 42%),** with a conservative finetuning recipe, tracing the published checkpoint's brittleness to recipe-induced overfitting and proposing an alternative baseline.
- **Doubled manipulation success on long-tail subtasks,** by oversampling skill transitions via boundary resampling.
- **Open-sourced a flow-matching VLA integration for RLinf,,** enabling RL training on the BEHAVIOR-1K suite in OmniGibson.
- **Running sim-to-real transfer of household tasks,** on a hand-built AlohaMini embodiment.

A deployed fleet fails in the long tail, and every result above started the same way: work out why the policy breaks, then fix it with data or recipe and measure against an honest baseline. I'd bring that habit to Helix's visuomotor policies, with behavior cloning, RL, and VLA reasoning as the toolkit, aimed at manipulation that holds up under real sensor noise and contact dynamics. It's the fastest way I know to iterate in real robotic environments.

I'm a Canadian citizen, TN-eligible with no sponsorship needed, and can relocate to San Jose within 2-4 weeks of an offer.